

|                                                                                                                                                    |                                                                                                                                                   |                                                                |                                                               |
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| <br><b>PT ETANA</b><br><b>BIOTECHNOLOGIES</b><br><b>INDONESIA</b> | <b>STANDARD OPERATING PROCEDURE</b>                                                                                                               |                                                                | Page : 1 of 8                                                 |
|                                                                                                                                                    | <b>TITLE :</b><br><b>PERAWATAN BERKALA TERHADAP MESIN LAF MOBILE TYPE ORABS</b><br><b>PREVENTIVE MAINTENANCE OF LAF MOBILE MACHINE TYPE ORABS</b> |                                                                | SOP No. : SOP/EBI/EN-065<br>Revision : 00<br>Effective Date : |
| <b>PREPARED BY :</b>                                                                                                                               | <b>REVIEWED BY :</b>                                                                                                                              | <b>REVIEWED BY :</b>                                           |                                                               |
| <b>PRIBADI SETIA BAYU AJI</b><br><b>(NON UTILITY ENGINEERING TECHNICIAN)</b>                                                                       | <b>WENDI RUKMANSYAH</b><br><b>(ENGINEERING SPV)</b>                                                                                               | <b>PURNO BUDI KISWANTO</b><br><b>(ENGINEERING SR. MANAGER)</b> |                                                               |
| <b>REVIEWED BY :</b>                                                                                                                               | <b>APPROVED BY :</b>                                                                                                                              | <b>APPROVED BY :</b>                                           |                                                               |
| <b>VINO SOADUON</b><br><b>(PROD SR. MANAGER)</b>                                                                                                   | <b>DEVIKA SRI WULANDARI</b><br><b>(QS MANAGER)</b>                                                                                                | <b>SHANDY</b><br><b>(QA MANAGER)</b>                           |                                                               |

## 1. TUJUAN

### PURPOSE

- 1.1. Memberikan petunjuk tentang perawatan terhadap Mesin *LAF Mobile Type ORABS (Open Restricted Access Barrier System)* agar sesuai dengan standar yang ada di PT Etana Biotechnologies Indonesia.  
*Provide instructions on the maintenance of the LAF Mobile Machine Type ORABS (Open Restricted Access Barrier System) to ensure compliance with existing standards at PT Etana Biotechnologies Indonesia.*
- 1.2. Menjaga dan memastikan bahwa *ORABS* selalu beroperasi dengan normal.  
*Maintain and ensure that the ORABS is always operating normally.*
- 1.3. Untuk memastikan kegiatan pemeliharaan rutin dan korektif terhadap sistem *ORABS* dilakukan secara efektif, terdokumentasi, dan sesuai standar guna menjamin keandalan operasional dan mencegah risiko kontaminasi dalam proses aseptik.  
*To ensure that routine and corrective maintenance activities on the ORABS system are carried out effectively, properly documented, and in compliance with applicable standards, in order to maintain operational reliability and prevent contamination risks in aseptic processes.*

## 2. RUANG LINGKUP

### SCOPE

SOP ini digunakan oleh departemen Engineering sebagai pedoman dalam melakukan perawatan terhadap Mesin *LAF Mobile Type ORABS (Open Restricted Access Barrier System)* yang ada di area produksi.

|                                                                                                                                                    |                                                                                                                                                                           |                                               |  |
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*This SOP is used by the Engineering department as a guideline in carrying out maintenance on the LAF Mobile Type ORABS (Open Restricted Access Barrier System) machines in the production area.*

### **3. TANGGUNG JAWAB RESPONSIBILITY**

#### **3.1 Teknisi bertanggung jawab:**

*Technician are responsible:*

3.1.1. Melakukan perawatan sesuai jadwal yang telah ditetapkan.

*Perform maintenance according to a predefined schedule.*

3.1.2. Menindaklanjuti Permintaan Jasa Engineering (PJE) dari departemen lain terkait masalah ORABS.

*Follow up Engineering Services Requests from other departments related ORABS issue.*

3.1.3. Melaporkan setiap ada penyimpangan kepada Supervisor Engineering.

*Report any deviation to Engineering Supervisor.*

#### **3.2 Supervisor Engineering bertanggung jawab:**

*Engineering Supervisor is responsible:*

3.2.1. Membuat jadwal perawatan secara berkala.

*Create schedule of maintenance periodically.*

3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.

*Ensure implementation of maintenance is according to the schedule.*

3.2.3. Mengadakan *review* bulanan terhadap hasil kerja teknisi.

*Conduct a monthly review to works of technicians.*

3.2.4. Memberikan training kepada departemen lain yang terkait.

*Provide training to the related department.*

3.2.5. Mengkoordinasikan dan melaporkan kepada Manager Engineering jika terdapat kerusakan pada sistem ORABS.

*Coordinate and report to Engineering Manager if there is malfunction to ORABS System.*

#### **3.3 Supervisor Kalibrasi & Validasi bertanggung jawab:**

*Calibration & Validation supervisor is responsible:*

Melakukan validasi dan kalibrasi alat ukur yang dipergunakan pada sistem ORABS seperti *magnehelic*.

|                                                                                                                                                    |                                                                                                                                                                           |                                               |
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*Performs validation and calibration of measuring instruments used on ORABS system like magnehelic.*

#### 3.4. Manajer Engineering bertanggung jawab:

*Engineering Manager is responsible:*

3.4.1. Menentukan suku cadang pengganti apabila suku cadang yang sesuai sudah tidak dijual lagi dengan pembuatan usulan perubahan apabila ada perubahan parts yang digunakan *refer to point 5.7.*

*Determine replacement parts if the appropriate parts are no longer available for sale by proposing changes if there are changes to the parts used. Refer to point 5.7.*

3.4.2. Memutuskan dan mengkoordinasi tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem ORABS.

*Decide and coordinate action required, if necessary replacement, equipment modification, change system of ORABS System.*

## 4. DEFINISI

### DEFINITION

#### 4.1. Pemeliharaan Perawatan

*Preventive Maintenance*

Pemeliharaan terjadwal yang dilakukan secara berkala untuk mencegah kerusakan atau penurunan kinerja sistem Aktivitas. Termasuk inspeksi, pembersihan, kalibrasi, dan penggantian komponen.

*Periodic maintenance carried out according to a defined schedule to prevent equipment malfunction or deterioration in performance. Activities include inspection, cleaning, calibration, and replacement of parts.*

#### 4.2. Kalibrasi

*Calibration*

Proses verifikasi dan penyesuaian alat ukur agar hasil pengukuran sesuai dengan standar yang telah ditentukan.

*The process of verifying and adjusting measuring instruments so that the measurement results comply with predetermined standards.*

#### 4.3. LAF (Laminar Air Flow)

*LAF (Laminar Air Flow)*

Sistem aliran udara yang bergerak secara seragam dan berkecepatan rendah, dalam arah yang sejajar dan sejajar terhadap permukaan kerja, dengan tujuan menjaga area tersebut tetap steril dari kontaminasi partikel udara.

|                                                                                                                                                    |                                                                                                                                                                           |                                               |
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*A low-speed, uniform airflow system, parallel and perpendicular to the work surface, designed to keep the area free from airborne particle contamination.*

#### 4.4. *ORABS (Open Restricted Access Barrier System)*

*ORABS (Open Restricted Access Barrier System)*

Sistem penghalang terbuka dengan aliran udara laminar (LAF) dan HEPA filter yang melindungi area kritis dari kontaminasi.

*An open barrier system with laminar airflow (LAF) and HEPA filters that protects critical areas from contamination.*

#### 4.5. *HEPA (High Efficiency Particulate Air filter)*

*HEPA (High Efficiency Particulate Air filter)*

Filter udara dengan partikulat efisiensi yang tinggi. Filter jenis ini baik untuk digunakan sebagai pembersih atau implementasi ruang-ruang dengan pengkondisian udara khusus.

*High efficiency particulate air filter. This type of filter is suitable for use as a cleaner or for implementation in rooms with special air conditioning.*

#### 4.6. *UPS (Uninterruptible Power Supplies)*

*UPS (Uninterruptible Power Supplies)*

Perangkat elektronik yang mampu menyediakan cadangan listrik sementara ketika arus listrik utama terputus.

*Electronic devices capable of providing temporary backup power when the main power supply is interrupted.*

## 5. **PETUNJUK UMUM**

### **GENERAL INSTRUCTION**

#### 5.1. Cara perawatan harus dilakukan dengan teratur dan sesuai dengan jadwal.

*The way of maintenance should be done regularly and in accordance with the schedule.*

#### 5.2. Teknisi yang telah terlatih atau sudah mendapatkan pelatihan dan terqualifikasi yang dapat melakukan perawatan sistem ORABS.

*Trained or qualified training technicians who can conduct maintenance of ORABS system.*

#### 5.3. Keselamatan kerja harus tetap di perhatikan selama perawatan.

*Work safety must be kept in mind during the maintenance.*

#### 5.4. Kegiatan perawatan yang dilakukan secara berkala harus didokumentasikan setelah perawatan dilakukan.

*Regular maintenance activities should be documented after maintenance done.*

#### 5.5. Apabila alat ukur tersebut didapati dalam keadaan rusak, harus diberi penandaan yang jelas.

|                                                                                                                                                    |                                                                                                                                                                           |                                               |
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*If the measuring instrument is found to be in a defective state, it should be clearly marked.*

- 5.6. Perlengkapan kerja harus sesuai dengan yang telah ditetapkan.

*Work equipment must be in accordance with the set.*

- 5.7. Penggantian *spare part* harus sesuai dengan spesifikasi dan fungsinya.

*Spare parts replacement must be in accordance with the specifications and functions.*

- 5.8. Untuk perubahan major yang mendasar dan merubah sistem *ORABS* harus memakai formulir *Change Control* dan disetujui oleh manajer terkait.

*For major underlying changes and changing the ORABS system must use a change control form and be approved by related managers.*

## 6. ALAT DAN BAHAN

### TOOLS AND MATERIAL

#### 6.1. ALAT

##### TOOLS

- 6.1.1. Kunci Inggris

*Wrench Key*

- 6.1.2. Obeng

*Screwdriver*

- 6.1.3. Kunci L

*Allen Key*

#### 6.2. BAHAN

##### MATERIAL

- 6.2.1. Ethanol 70%

*Ethanol 70%*

- 6.2.2. Kain Wipes

*Wipes fabric*

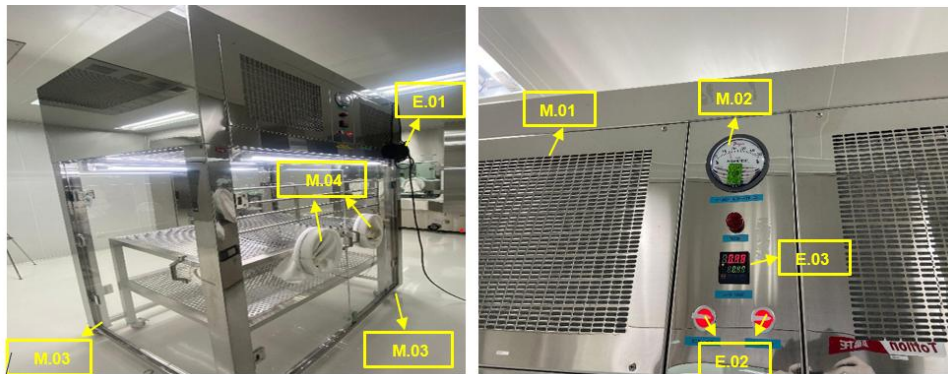
## 7. PROSEDUR

### PROCEDURE

- 7.1. Perawatan 3 bulanan (L2) pada *Open Restricted Access Barrier System (ORABS)*.

*Three (3) month maintenance 3 month (L2) of Open Restricted Access Barrier System (ORABS).*

|                                                                                                                                                    |                                                                                                                                                                           |                                               |
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*M = Mechanical*

*E = Electrical*

Gambar 1. Mesin LAF Mobile Type ORABS (Open Restricted Access Barrier System)

Figure 1. LAF Mobile Machine Type ORABS (Open Restricted Access Barrier System)

- 7.1.1. Cek motor *blower* dari suara tidak normal, perbaiki atau ganti bila perlu. Lihat pada gambar 1 kode M01.

*Check the blower motor from abnormal sound, repair or replace if necessary. Look at the figure 1 code M01.*

- 7.1.2. Cek koneksi power 1 phasa, perbaiki jika koneksi power terlihat rusak. Lihat pada gambar 1 kode E01.

*Check the single-phase power connection and repair any if visible damage. Look at the figure 1 code E.01.*

- 7.1.3. Cek perbedaan tekanan HEPA filter. Jika kurang dari 100 pascal artinya ada kebocoran HEPA, seal HEPA atau selang *Magnahelic*. Jika lebih dari 500 pascal artinya mampet, dan segera di perbaiki. Jika sudah diperbaiki tekanan HEPA masih belum masuk syarat (Syarat *Pressure Magnahelic* 200 – 300 pascal), segera infokan ke team produksi untuk meminta waktu melakukan penggantian HEPA Filternya. Penggantian HEPA filter maksimal 5 tahun jika hasil integrity masih masuk syarat (syarat integrity < 0.005%). Lihat pada gambar 1 kode M02.

*Check the pressure difference of the HEPA filter. If it is less than 100 pascals, it means there is a leak in the HEPA, HEPA seal, or Magnahelic hose. If it is more than 500 pascals, it means it is clogged, and must be repaired immediately. If the HEPA pressure still does not meet the requirements (Magnahelic Pressure Requirements 200–300 pascal) after repair, immediately inform the production team to request time to replace the HEPA filter. HEPA filters should be replaced every 5 years at most if the integrity results still meet the requirements (integrity requirements < 0.005%). Look at the figure 1 code M02.*

|                                                                                                                                                    |                                                                                                                                                                           |                                               |  |
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- 7.1.4. Periksa dan kencangkan semua koneksi terminal kabel, *circuit breaker*, *relay*, ganti jika terjadi perubahan warna pada komponen *part electricalnya*. Lihat pada gambar 1 kode E02.

*Check and tighten all cable terminal connections, circuit breakers, relays, replace if there is a color change on the electrical components. Look at the figure 1 code E02.*

- 7.1.5. Periksa kondisi display controller, perbaiki atau ganti jika terlihat rusak. Lihat pada gambar 1 kode E03.

*Check the condition of the display controller, repair or replace it if it looks damaged. Look at figure 1 code E03.*

- 7.1.6. Cek baut roda, kencangkan jika kendur dan ganti rodanya jika terlihat rusak. Lihat pada gambar 1 kode M03.

*Check the wheel bolts, tighten them if they are loose and replace the wheel if they appear damaged. Look at the figure 1 code M03.*

- 7.1.7. Cek kondisi sarung tangan untuk ORABS. Fungsi kebocoran sarung tangan dilakukan oleh team produksi saat proses leak test glove menggunakan *Glove Leak Test (GLT)*. Lihat pada gambar 1 kode M04.

*Check the condition of gloves for ORABS. Glove leakage testing is performed by the production team during the glove leak test process using glove leak test (GLT). Look at the figure 1 code M04.*

- 7.1.8. Periksa kekencangan baut2 pada pengikat cover – cover di mesin.

*Check the tightness of the bolts on the cover fasteners on the machine.*

- 7.1.9. Periksa kondisi kaca akrilik, jika terdapat yang terlihat retak / pecah segera infokan ke supervisor untuk di perbaiki atau diganti dengan yang baru.

*Check the condition of the acrylic glass. If there are any visible cracks or breaks, immediately inform the supervisor so that it can be repaired or replaced with a new one.*

## 8. REFERENSI

### REFERENCE

Petunjuk SOP/EBI/FF-027 Pengoperasian dan Pembersihan Open Restricted Access Barrier System (ORABS) Mobile.

*SOP/EBI/FF-027 Instructions for Operating and Cleaning the Mobile Open Restricted Access Barrier System (ORABS).*



|                                                                                                                                                    |                                                                                                                                                                           |                                               |
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## 9. LAMPIRAN

### ATTACHMENT

Formulir No. SOP/EBI/EN-065-F01A Perawatan Berkala Terhadap Mesin *LAF Mobile Type ORABS*.

*Form No. SOP/EBI/EN-065-F01A Periodic Maintenance of LAF Mobile Type ORABS Machines.*

## 10. RIWAYAT

### HISTORY

| <b>Tanggal</b><br><i>Date</i> | <b>No. Change Control</b><br><i>Change Control No.</i> | <b>No. Revisi</b><br><i>No. Revision</i> | <b>Alasan Perubahan</b><br><i>Reason of Change</i> |
|-------------------------------|--------------------------------------------------------|------------------------------------------|----------------------------------------------------|
|                               | CC-EBI-EN-015-25                                       | 00                                       | Dokumen baru<br><i>New Document</i>                |

## 11. DOKUMEN TERKAIT

### RELATED DOCUMENT

Tidak ada

N/A

## 12. TARGET PELATIHAN

### TRAINING TARGET

| <b>Departemen</b><br><i>Department</i>                    | <b>Jabatan Personil</b><br><i>Personil Title</i> |
|-----------------------------------------------------------|--------------------------------------------------|
| Departement Engineering<br><i>Engineering Departement</i> | Teknisi<br><i>Technician</i>                     |

## 13. DISTRIBUSI

### DISTRIBUTION

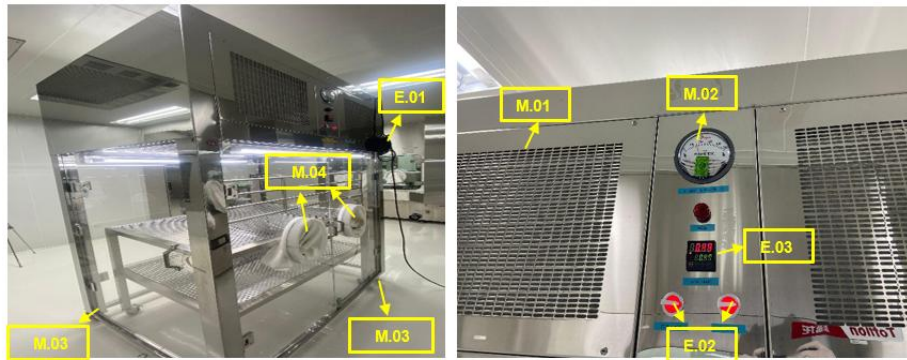
- 13.1. Departemen Engineering (*Hard copy*).  
*Engineering Department (Hard copy).*
  - 13.2. Departemen Sistem Mutu (*Hard copy*).  
*Quality System Departement (Hard copy).*
  - 13.3. Departemen Produksi (*Hard copy*).  
*Production Department (Hard copy).*
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| <br><b>PT ETANA</b><br><b>BIOTECHNOLOGIES</b><br><b>INDONESIA</b> | <b>FORMULIR / FORM</b>                                                                                                                                  | Document No. :<br>SOP/EBI/EN-065-F01A |
|                                                                                                                                                    | <b>PERAWATAN BERKALA TERHADAP</b><br><b>MESIN LAF MOBILE TYPE ORABS</b><br><br><b>PREVENTIVE MAINTENANCE OF LAF</b><br><b>MOBILE MACHINE TYPE ORABS</b> |                                       |

|             |   |  |
|-------------|---|--|
| Level       | : |  |
| Month       | : |  |
| Date        | : |  |
| Day         | : |  |
| Hours Meter | : |  |

| Level | Component | Activity |   |   |   |   |   |   |   | Condition |  | Remark |
|-------|-----------|----------|---|---|---|---|---|---|---|-----------|--|--------|
|       |           | C        | I | L | R | S | T | W | B | G         |  |        |



|           |                                                     |  |  |  |  |  |  |  |  |  |  |  |
|-----------|-----------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|
| <b>L2</b> |                                                     |  |  |  |  |  |  |  |  |  |  |  |
|           | Blower motor, M01                                   |  |  |  |  |  |  |  |  |  |  |  |
|           | Connection power 1 phasa, E01                       |  |  |  |  |  |  |  |  |  |  |  |
|           | Pressure magnehelic (Requirement 200 – 300 Pa), M02 |  |  |  |  |  |  |  |  |  |  |  |
|           | Connection terminal, circuit braker and relay, E02  |  |  |  |  |  |  |  |  |  |  |  |
|           | Function Display controller, E03                    |  |  |  |  |  |  |  |  |  |  |  |
|           | Condition wheel, M03                                |  |  |  |  |  |  |  |  |  |  |  |
|           | Gloves for ORABS, M04                               |  |  |  |  |  |  |  |  |  |  |  |
|           | tightness of the bolts                              |  |  |  |  |  |  |  |  |  |  |  |
|           | condition of the acrylic glass                      |  |  |  |  |  |  |  |  |  |  |  |

|        |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|--|--|--|--|--|--|--|--|--|--|--|--|
| Note : |  |  |  |  |  |  |  |  |  |  |  |  |
|--------|--|--|--|--|--|--|--|--|--|--|--|--|

| No | Date | Start | Finish | Duration |        | Total personnel | Area | Remark |
|----|------|-------|--------|----------|--------|-----------------|------|--------|
|    |      |       |        | Hour     | Minute |                 |      |        |
|    |      |       |        |          |        |                 |      |        |
|    |      |       |        |          |        |                 |      |        |

|                                                                                                                                                                                                                                                                          |                                   |                                           |                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|----------------------------------------------|
| Remark :<br>C : Cleaning (pembersihan)<br>I : Inspection (pemeriksaan)<br>L : Lubrication (pelumasan)<br>R : Replacement (penggantian)<br>S : Setting (penyetelan)<br>T : Tightened (dikencangkan)<br>W : Welding (pengelasan)<br>G : Good ( baik )<br>B : Bad ( jelek ) | Level :<br><br>L2 : Every 3 month | Done by :<br><br>(                      ) | Checked by :<br><br>(                      ) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|----------------------------------------------|

